

Classical Electrodynamics

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These notes are not endorsed by the lecturers. All errors are mine.

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0. ...

1. Electrostatics

1.1. Coulomb's law

We start our discussion of electrostatics with the hopefully familiar Coulomb's law¹

$$\mathbf{F} = kq_1q_2 \frac{\mathbf{x}_1 - \mathbf{x}_2}{|\mathbf{x}_1 - \mathbf{x}_2|^3}, \quad (1.1)$$

which expresses the force on a point charge q_1 sitting at $\mathbf{x}_1$ due to another point charge q_2 sitting at $\mathbf{x}_2$. Here,

$$k = \frac{1}{4\pi\epsilon_0} \simeq 8.99 \cdot 10^9 \frac{\text{Nm}^2}{\text{C}^2},$$

is known as Coulomb's constant, and is related to

$$\epsilon_0 \simeq 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{Nm}^2},$$

which is the absolute permittivity of vacuum.

Consider a test charge q . By this we mean q is infinitesimally small, meaning its presence has no effect on the field configuration sourced by other charges.² We define the electric field at the position $\mathbf{x}$ occupied by q as

$$\mathbf{F} = q\mathbf{E}.$$

Taking $\mathbf{E}$ to be sourced by a single charge q_0 placed at $\mathbf{x}_0$, we have by Coulomb's law

$$\mathbf{E}(\mathbf{x}) = kq_0 \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|^3}.$$

With multiple sources electric fields add as vectors.³ A discrete set of sources q_i placed at $\mathbf{x}_i$ produces an electric field

$$\mathbf{E}(\mathbf{x}) = k \sum_i q_i \frac{\mathbf{x} - \mathbf{x}_i}{|\mathbf{x} - \mathbf{x}_i|^3}.$$

Likewise, a continuous charge distribution with density $\rho(\mathbf{x})$ produces an electric field

$$\mathbf{E}(\mathbf{x}) = k \int_{\mathbb{R}^3} \rho(\mathbf{x}') \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3} d^3x'. \quad (1.2)$$

These are of course the same when $\rho(\mathbf{x}') \sim \sum_i q_i \delta^{(3)}(\mathbf{x}' - \mathbf{x}_i)$.

¹Experimental fact!

²This is the probe approximation.

³Experimental fact!

1.2. Gauss' law

Applying Coulomb's law we can derive Gauss' law.

Let S be an oriented surface in $\mathbb{R}^3$, and denote the unit normal to S by $\mathbf{n}$. Consider an infinitesimal piece σ of S , along which $\mathbf{n}$ is approximately constant, and denote the area element by da .⁴ We define the infinitesimal flux (through σ) of the electric field as

$$\varphi_\sigma(\mathbf{E}) \equiv \mathbf{E} \cdot \mathbf{n} da.$$

Now, suppose the electric field is sourced by a single point particle of charge q , and let $\mathbf{r}$ be the distance between the source and σ . Then

$$\varphi_\sigma(\mathbf{E}) \stackrel{\text{Coulomb's}}{=} kq \frac{\mathbf{r} \cdot \mathbf{n}}{|\mathbf{r}|^3} da = kq \frac{\cos \theta}{|\mathbf{r}|^2} da = kq \frac{da_\perp}{|\mathbf{r}|^2} = kq d\Omega,$$

where θ is the angle between $\mathbf{E}$ and $\mathbf{n}$, we also used $da_\perp = |\mathbf{r}|^2 d\Omega$.⁵ Here, $d\Omega = \sin \theta' d\theta' d\phi'$ is the solid angle subtended by da as seen by an observer sitting at the point charge.

The total flux is found by integrating over S . Hence, over the solid angle as measured in coordinates centered at q . When S encloses the charge q , the total solid angle is 4π . However, if q is outside S the total solid angle is 0. The total flux is then

$$\varphi_S(\mathbf{E}) = \oint_S \mathbf{E} \cdot \mathbf{n} da = \begin{cases} 4\pi kq & q \text{ enclosed,} \\ 0 & q \text{ outside.} \end{cases}$$

For a discrete set of charges, the right-hand side becomes a sum over all charges enclosed by S . For a continuous charge distribution with density $\rho(\mathbf{x}')$ we have

$$\varphi_S(\mathbf{E}) = \oint_S \mathbf{E} \cdot \mathbf{n} da = \frac{1}{\epsilon_0} \int_V \rho(\mathbf{x}) d^3x, \quad (1.3)$$

where V is a volume such that $\partial V = S$. This is the integral form of Gauss' law.

Using the divergence theorem

$$\oint_S \mathbf{A} \cdot \mathbf{n} da = \int_V \nabla \cdot \mathbf{A} d^3x,$$

we can rewrite the surface integral

$$\oint_S \mathbf{E} \cdot \mathbf{n} da = \int_V \nabla \cdot \mathbf{E} d^3x,$$

⁴ $\mathbf{n}$ is normal to da .

⁵Only the transverse area $da_\perp$ contributes to $d\Omega$ and $da_{\text{sphere}} \propto r^2$.

giving

$$\int_V \nabla \cdot \mathbf{E} \, d^3x = \frac{1}{\epsilon_0} \int_V \rho(\mathbf{x}) \, d^3x.$$

Since the volume V is arbitrary we obtain the differential form of Gauss' law

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (1.4)$$

This equation can be used to compute the electric field from a known charge distribution. Thus we have found an alternative to (1.2). However, Gauss' law provides a solution up to addition of some divergenceless background field $\mathbf{E} \sim \mathbf{E} + \mathbf{E}_0$.⁶ Such ambiguities are commonplace in physics and should be familiar.

1.3. The scalar potential Φ

The electric field $\mathbf{E}(\mathbf{x})$ is a vector field, which is specified by three functions $E_i(\mathbf{x})$. This is ~~sometimes~~ annoying, hence we switch to a description in terms of a single function, the scalar potential $\Phi(\mathbf{x})$

$$\mathbf{E}(\mathbf{x}) = -\nabla\Phi(\mathbf{x}). \quad (1.5)$$

This being possible is non-obvious. However, in electrostatics (important!) this follows from Coulomb's law since

$$\nabla \frac{1}{|\mathbf{x} - \mathbf{x}'|} = -\frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3},$$

meaning we can write

$$\mathbf{E}(\mathbf{x}) = -k\nabla \int_{\mathbb{R}^3} \rho(\mathbf{x}') \frac{1}{|\mathbf{x} - \mathbf{x}'|} \, d^3x',$$

which implies

$$\Phi(\mathbf{x}) = k \int_{\mathbb{R}^3} \rho(\mathbf{x}') \frac{1}{|\mathbf{x} - \mathbf{x}'|} \, d^3x'. \quad (1.6)$$

This reduces to the familiar Coulomb potential for a single pointlike source placed at $\mathbf{x}_0$

$$\Phi(\mathbf{x}) = kq_0 \frac{1}{|\mathbf{x} - \mathbf{x}_0|}.$$

The existence of the scalar potential immediately implies $\mathbf{E}$ has zero curl⁷

⁶Specified by boundary conditions.

⁷Since $\nabla \times (\nabla f) = 0$.

$$\nabla \times \mathbf{E} = 0. \quad (1.7)$$

The scalar potential is related to the electrostatic potential energy of a pointlike particle with charge q in the electric field. The well-defined quantity is the difference between potential energies of the charged particle, at two points $\mathbf{x}_1$ and $\mathbf{x}_2$. The difference corresponds to the work done by us (opposite the work done by $\mathbf{E}$) to move the particle from $\mathbf{x}_1$ to $\mathbf{x}_2$ along some path

$$W = \overset{\text{since opposite}}{\overline{=}} \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{F} \cdot d\mathbf{l} = -q \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{E} \cdot d\mathbf{l} = q \int_{\mathbf{x}_1}^{\mathbf{x}_2} \nabla \Phi \cdot d\mathbf{l} \overset{\text{by definition}}{=} q \int_{\mathbf{x}_1}^{\mathbf{x}_2} d\Phi.$$

This is a position-dependent quantity associated to some probe charge, which stores the energy we put into moving it around in the electric field. This means we can interpret $q\Phi$ as the potential energy of the charged particle in the field $\mathbf{E}$

$$U(\mathbf{x}) = q\Phi(\mathbf{x}).$$

Since moving a probe charge in an electric field costs energy, then creating a given charge configuration costs energy. The work done is the total potential energy of the charge configuration. This can be computed by summing the work needed to carry a charge q_i from infinity to its position $\mathbf{x}_i$ in the final configuration, one-by-one

$$U = k \sum_{i=1}^n \sum_{\substack{j < i \\ q_i \Phi(\text{from } q_j)}} \frac{q_i q_j}{|\mathbf{x}_i - \mathbf{x}_j|} = \frac{k}{2} \sum_{i \neq j}^n \frac{q_i q_j}{|\mathbf{x}_i - \mathbf{x}_j|}.$$

The potential energy of a continuous charge distribution is

$$U = \frac{k}{2} \iint \frac{\rho(\mathbf{x})\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x d^3x', \quad (1.8)$$

where regularisation is required to avoid divergences.⁸ This expression can be rewritten as

$$U = \frac{1}{2} \int \rho(\mathbf{x})\Phi(\mathbf{x}) d^3x$$

$$\overset{\text{Gauss' law}}{=} \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E})\Phi d^3x$$

⁸From integration along $\mathbf{x} = \mathbf{x}'$, corresponding to the infinite self-energy of a charge in its own potential.

$$\begin{aligned}
&= -\frac{\epsilon_0}{2} \int (\nabla^2 \Phi) \Phi \, d^3x \\
&\stackrel{(9)}{=} \frac{\epsilon_0}{2} \int (\nabla \Phi) \cdot (\nabla \Phi) \, d^3x \\
&= \frac{\epsilon_0}{2} \int |\mathbf{E}|^2 \, d^3x.
\end{aligned}$$

This tells us $|\mathbf{E}|^2$ can be thought of as the electrostatic energy density!

Going back to the definition of work we find an expression for a finite variation of the scalar potential along a path $\mathcal{P}$

$$\Delta \Phi = - \int_{\mathcal{P}} \mathbf{E} \cdot d\mathbf{l}.$$

Up to an overall shift by a constant $\Phi \rightarrow \Phi + c$, the scalar potential is a single-valued function of $\mathbf{x}$. This means the variation must vanish for closed $\mathcal{P}$

$$0 = \oint_{\mathcal{P}} \mathbf{E} \cdot d\mathbf{l}.$$

Using Stokes' theorem

$$\varphi_S(\nabla \times \mathbf{A}) = \int_S (\nabla \times \mathbf{A}) \cdot \mathbf{n} \, da = \oint_{\gamma} \mathbf{A} \cdot d\mathbf{l},$$

this implies the flux vanishes

$$0 = \oint_S (\nabla \times \mathbf{E}) \cdot \mathbf{n} \, da,$$

integrated over an arbitrary surface S with boundary $\partial S = \mathcal{P}$. This is comforting since we know $\nabla \times \mathbf{E} = 0$.¹⁰

Gauss' law can also be expressed in terms of the scalar potential

$$\nabla^2 \Phi(\mathbf{x}) = -\frac{\rho(\mathbf{x})}{\epsilon_0}, \tag{1.9}$$

which is the Poisson equation. This becomes the Laplace equation when there are no sources

$$\nabla^2 \Phi(\mathbf{x}) = 0. \tag{1.10}$$

⁹Consider $\nabla \cdot (\Phi \nabla \Phi) = (\nabla \Phi) \cdot (\nabla \Phi) + \Phi \nabla^2 \Phi$ and use the divergence theorem.

¹⁰This leads credence to the scalar potential being single-valued.

1.4. Green's functions

We have already seen some ways to compute Φ . However, none of these know how to handle things like discontinuities or boundary conditions (which is a problem!).

So we need more general tools to describe the scalar potential. One way to proceed is based on Green's theorem, which given two arbitrary functions ϕ and ψ says¹¹

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3x = \oint_S \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) da.$$

Applying this with $\phi = \Phi(\mathbf{x}')$ and $\psi = 1/|\mathbf{x} - \mathbf{x}'|$ we obtain¹²

$$\begin{aligned} \int_V \left(\Phi(\mathbf{x}') \nabla'^2 \frac{1}{|\mathbf{x} - \mathbf{x}'|} - \frac{1}{|\mathbf{x} - \mathbf{x}'|} \nabla'^2 \Phi(\mathbf{x}') \right) d^3x' \\ = \oint_S \left(\Phi(\mathbf{x}') \frac{\partial}{\partial n'} \frac{1}{|\mathbf{x} - \mathbf{x}'|} - \frac{1}{|\mathbf{x} - \mathbf{x}'|} \frac{\partial \Phi(\mathbf{x}')}{\partial n'} \right) da' \end{aligned}$$

Now, using the Poisson equation, the identity

$$\nabla'^2 \frac{1}{|\mathbf{x} - \mathbf{x}'|} = -4\pi \delta^{(3)}(\mathbf{x} - \mathbf{x}'),$$

and taking $\mathbf{x} \in V$, the left-hand side becomes

$$\int_V \left(-4\pi \Phi(\mathbf{x}') \delta(\mathbf{x} - \mathbf{x}') + \frac{\rho(\mathbf{x}')}{\epsilon_0 |\mathbf{x} - \mathbf{x}'|} \right) d^3x' = \underbrace{-4\pi \Phi(\mathbf{x})}_{\text{vanishes when } \mathbf{x} \notin V} + \int_V \frac{\rho(\mathbf{x}')}{\epsilon_0 |\mathbf{x} - \mathbf{x}'|} d^3x'.$$

So we obtain an equation for the scalar potential¹³

$$\begin{aligned} \Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\ + \frac{1}{4\pi} \oint_S \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \frac{\partial \Phi(\mathbf{x}')}{\partial n'} - \Phi(\mathbf{x}') \frac{\partial}{\partial n'} \frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) da'. \end{aligned} \quad (1.11)$$

This expression says we can compute $\Phi(\mathbf{x})$ if we know

- (i) The charge distribution within a given volume V .
- (ii) The scalar potential on the surface S bounding V .

¹¹The normal derivative is defined as

$$\frac{\partial f}{\partial n} = \mathbf{n} \cdot \nabla f.$$

Where $\mathbf{n}$ is the normal to S at $\mathbf{x}$. The theorem can be derived by applying the divergence theorem to $\mathbf{A} = \phi \nabla \psi$.

¹²Here, $\mathbf{x}$ is the “observation-point”, i.e. we want an expression for $\Phi(\mathbf{x})$.

¹³Taking S to infinity we the surface integral vanishes.

(iii) The normal electric field on the surface S bounding V .

The latter two are needed to account for the effect of all charges lying outside V . So we only need to know the charge distribution inside V to compute the scalar potential, as long as we know the boundary behaviour!¹⁴

When $\rho(\mathbf{x}') = 0$, meaning there are no charges in V , then the scalar potential satisfies the Laplace equation in V . When this is the case, our equation says the scalar potential is fully determined by the boundary behaviour (specifying Φ and $\partial\Phi/\partial n$ on S is known as a Cauchy boundary condition). This turns out to overconstrain the Laplace equation. So our equation does not express $\Phi(\mathbf{x})$ in terms of the boundary values, but only expresses a relation between several pieces of a consistent solution to the Poisson equation. Then, when supplying the values of Φ and its normal derivative on S , one needs to make sure these are compatible, meaning they correspond to the same solution to the Poisson equation. This makes the equation less useful.

However, in order to specify a unique solution to the Poisson equation, we only need to specify either boundary conditions for the potential (known as Dirichlet boundary conditions), or for its normal derivative (known as Neumann boundary conditions).¹⁵ Our equation would then become immensely more useful if it involves Dirichlet or Neumann boundary conditions (not both!).

This can be solved by using the Green's function for the Laplacian

$$\nabla'^2 G(\mathbf{x}, \mathbf{x}') = -4\pi\delta^{(3)}(\mathbf{x} - \mathbf{x}'). \quad (1.12)$$

Now, when deriving our equation for the scalar potential the important property of ψ we needed was

$$\nabla'^2 \psi = -4\pi\delta^{(3)}(\mathbf{x} - \mathbf{x}').$$

So the derivation is completely analogous if we replaced ψ by the Green's function. However, the Green's function works up to addition by an arbitrary function which solves the Laplace equation

$$G(\mathbf{x}, \mathbf{x}') \sim G(\mathbf{x}, \mathbf{x}') + F(\mathbf{x}, \mathbf{x}'), \quad \nabla'^2 F(\mathbf{x}, \mathbf{x}') = 0.$$

¹⁴This is wild since V is completely arbitrary.

¹⁵This can be shown using Green's theorem, see Jackson.

This is nice since we can pick $F(\mathbf{x}, \mathbf{x}')$ to remove either $\Phi(\mathbf{x}')$ or $\partial\Phi(\mathbf{x}')/\partial n'$ at S !¹⁶

We consider

$$\begin{aligned}\Phi(\mathbf{x}) &= \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G(\mathbf{x}, \mathbf{x}') d^3x' \\ &+ \frac{1}{4\pi} \oint_S \left(G(\mathbf{x}, \mathbf{x}') \frac{\partial\Phi(\mathbf{x}')}{\partial n'} - \Phi(\mathbf{x}') \frac{\partial}{\partial n'} G(\mathbf{x}, \mathbf{x}') \right) da'.\end{aligned}$$

Picking $F(\mathbf{x}, \mathbf{x}')$ such that

$$G_D(\mathbf{x}, \mathbf{x}') = 0, \text{ for } \mathbf{x}' \in S \quad (1.13)$$

we find an equation for the scalar potential only in terms of Φ on S (Dirichlet boundary condition)

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G_D(\mathbf{x}, \mathbf{x}') d^3x' - \frac{1}{4\pi} \oint_S \Phi(\mathbf{x}') \frac{\partial}{\partial n'} G_D(\mathbf{x}, \mathbf{x}') da'. \quad (1.14)$$

While picking $F(\mathbf{x}, \mathbf{x}')$ such that¹⁷

$$\frac{\partial G_N}{\partial n'}(\mathbf{x}, \mathbf{x}') = -\frac{4\pi}{A_S}, \text{ for } \mathbf{x}' \in S, \quad (1.15)$$

where A_S is the area of S , we find an equation for the scalar potential only in terms of $\partial\Phi/\partial n'$ on S (Neumann boundary condition), and the average of the potential on S

$$\begin{aligned}\Phi(\mathbf{x}) &= \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G_N(\mathbf{x}, \mathbf{x}') d^3x' \\ &+ \langle\Phi\rangle_S + \frac{1}{4\pi} \oint_S G_N(\mathbf{x}, \mathbf{x}') \frac{\partial\Phi(\mathbf{x}')}{\partial n'} da'.\end{aligned} \quad (1.16)$$

However, since the scalar potential is only defined up to a constant the term $\langle\Phi\rangle_S$ is meaningless, so we simply ignore it.

The main problem now is of course determining $G(\mathbf{x}, \mathbf{x}')$.

¹⁶These $F(\mathbf{x}, \mathbf{x}')$ represent the contribution to the scalar potential inside V due to sources outside V .

¹⁷Picking

$$\frac{\partial G_N}{\partial n'}(\mathbf{x}, \mathbf{x}') = 0,$$

leads to issues, immediate by Gauss' theorem.

2. Finding Φ and boundary-value problems

2.1. Electric fields across surface distributions

Let S be a surface with normal $\mathbf{n}$ directed from “side 1” toward “side 2”¹⁸, and suppose there is a charge density $\sigma(\mathbf{x})$ on the surface. By this we mean an area δA carries charge δQ given by

$$\delta Q = \int_{\delta A} \sigma(\mathbf{x}) d^2x.$$

Now, consider a small cylinder C with circular area δA . The cylinder has a small enough height, such that the two circular faces lie on opposite sides of S .

Applying Gauss’ law one obtains

$$\varphi_C(\mathbf{E}) = \oint_C \mathbf{E} \cdot \mathbf{n} da \simeq \delta A (\mathbf{E}_2 \cdot \mathbf{n} - \mathbf{E}_1 \cdot \mathbf{n}) = \frac{\delta Q}{\epsilon_0} = \frac{\sigma \delta A}{\epsilon_0}.$$

We only include contributions from the circular faces, and take δA small enough, such that the fields $\mathbf{E}_i$ are approximately constant across the faces.¹⁹ Taking $\delta A \rightarrow dA$ we find an equation for the discontinuity of $E_\perp$ across surface distributions

$$(\mathbf{E}_2(\mathbf{x}) - \mathbf{E}_1(\mathbf{x})) \cdot \mathbf{n} = E_{2,\perp}(\mathbf{x}) - E_{1,\perp}(\mathbf{x}) = \frac{\sigma(\mathbf{x})}{\epsilon_0},$$

which holds for any $\mathbf{x} \in S$.

However, the parallel component of $\mathbf{E}$ is continuous across S . This can be seen by considering a small rectangle across S with two long sides, and negligible height.²⁰ Then single-valuedness of the scalar potential implies continuity of $E_\parallel$

$$0 \stackrel{\text{well-defined}}{=} \oint \mathbf{E} \cdot d\mathbf{l} \simeq (\mathbf{E}_2 - \mathbf{E}_1) \cdot \frac{\mathbf{t}L}{L},$$

where $\mathbf{t}$ is the unit tangent to the rectangle’s sides running parallel to S , and L is the length of the rectangle. We are free to change the orientation, meaning $\mathbf{t}$ is an arbitrary tangent to S . This implies the electric field along any direction tangent to S is continuous across S

$$\mathbf{E}_{2,\parallel} = \mathbf{E}_{1,\parallel}.$$

¹⁸Think of $\mathbf{n}$ as piercing the surface.

¹⁹The minus sign comes from $\mathbf{n}_1 = -\mathbf{n}_2$.

²⁰This rectangle “pops” out of S .

2.2. Conductors as Dirichlet boundaries

Conductors are materials capable of storing charge, inside which the electric field vanishes. This is the case since if $\mathbf{E} \neq 0$ there would be a force $\mathbf{F} = q\mathbf{E}$ on the charges inside the conductor, leading to a redistribution of charge (since charges can move in a conductor). This means any (small) electric field inside a conductor will immediately be counterbalanced by such a redistribution. An immediate consequence of this is all points are equipotential

$$\Phi(\mathbf{x}) = \text{constant.} \quad (\text{in conductors})$$

The presence of a conductor is therefore equivalent to the presence of a surface S where the potential has some fixed value. This makes conductors a typical example of Dirichlet boundary conditions.

The above immediately implies the electric field at the surface of a conductor has no component parallel to the surface, otherwise we would have a variation of the scalar potential $E_{\parallel} = -\nabla_{\parallel}\Phi \neq 0$, which is a contradiction. The electric field at the surface must therefore be normal to the surface

$$E_{\parallel} = 0, \quad \mathbf{E} = \mathbf{E}_{\perp}. \quad (\text{outside a conductor's surface})$$

Now, to compute the potential (in the presence of conductors) we only need to know the Green's function G_D ! This guy depends on the shape of our conductor, and can only be found on a case-by-case basis.

2.3. Method of images

The method of images is based on the idea that the presence of a conductor on a surface S can be “simulated” by some appropriate configuration of charges $\rho(\mathbf{x})$

$$\text{conductor on } S \rightsquigarrow \rho(\mathbf{x}) \text{ with } \Phi(S) = \text{constant.}$$

This allows one to replace the potentially annoying boundary condition for $\Phi(\mathbf{x})$ with new “imaginary” sources whose contribution to $\Phi(\mathbf{x})$ is easily computed.²¹ We will now cover some simple examples.

²¹e.g. by Coulomb's.

2.3.1. Grounded conducting sphere

Consider a spherical conductor centered at the origin with radius a , in presence of a pointlike charge q located at $\mathbf{y}$ with $|\mathbf{y}| > a$ (so q is outside). We assume the sphere is grounded, meaning

$$\Phi(|\mathbf{x}| = a) = 0.$$

To determine the potential elsewhere, we make the ansatz that the conducting sphere can be modeled by a single pointlike charge q' at some $\mathbf{y}'$. By symmetry we take $\mathbf{y}' = c\mathbf{y}$ for some $0 < c < 1$, meaning both lie along the same ray. Let $\mathbf{n}_q$ be the unit vector of this ray.

The potential would then be

$$\begin{aligned}\Phi(\mathbf{x}) &= \frac{q}{4\pi\epsilon_0} \frac{1}{|\mathbf{x} - \mathbf{y}|} + \frac{q'}{4\pi\epsilon_0} \frac{1}{|\mathbf{x} - \mathbf{y}'|} \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{x|\mathbf{n} - \frac{y}{x}\mathbf{n}_q|} + \frac{q'}{4\pi\epsilon_0} \frac{1}{y'|\frac{x}{y'}\mathbf{n} - \mathbf{n}_q|},\end{aligned}$$

where $\mathbf{n}$ is the unit vector for the observation point $\mathbf{x}$. Requiring this vanishes at $|\mathbf{x}| = a$ leads to

$$\frac{q}{a} = -\frac{q'}{y'}, \quad \frac{y}{a} = \frac{a}{y'},$$

or

$$q' = -\frac{a}{y}q, \quad y' = \frac{a^2}{y}.$$

With the potential found we can determine $\mathbf{E}$ by taking the gradient. Likewise, we can determine the surface charge density (since $\mathbf{E}_{\text{in}} = 0$ and $E_{\parallel} = 0$ on the conductor)

$$\sigma = \epsilon_0 \underbrace{\left(-\frac{\partial\Phi}{\partial n}\right)_{x=a}}_{\mathbf{E}_{\text{out}} \cdot \mathbf{n}} = -\frac{q}{4\pi a^2} \frac{q}{y} \frac{1 - a^2/y^2}{\left(1 + a^2/y^2 - 2\frac{a}{y}\cos\gamma\right)^{3/2}},$$

where γ is the angle between $\mathbf{x}$ (on the sphere) and the direction of $\mathbf{y}$ ($\mathbf{n} \cdot \mathbf{n}_q = \cos\gamma$).

- (i) We find q induces a charge distribution on the grounded sphere of opposite sign. We would expect this from Coulomb's law, since the charges of opposite sign would be the ones moving around in the conductor to come closer to q , until reaching equilibrium.

(ii) The total induced charge on the sphere is

$$\oint_S \sigma \, da = \oint_S \epsilon_0 \left(-\frac{\partial \Phi}{\partial n} \right)_{x=a} da = \oint_S \epsilon_0 \mathbf{E} \cdot \mathbf{n} \, da = \epsilon_0 \varphi_S(\mathbf{E}) \stackrel{\text{Gauss' law}}{=} q',$$

where we apply Gauss' law on the field configuration including the image charge.

(iii) The distribution peaks when $\gamma = 0$, meaning charges confined to the conductor try to be as close as possible to q . The spread of the peak is determined by the coefficient of $\cos \gamma$. When a/y is large the angular dependence becomes more important. This is the case when the charge is close to the conductor.

The charge inducing a charge distribution on the conductor leads to a backreaction on the charge. This occurs since the induced distribution produces its own electric field $\mathbf{E}_\sigma$, which exerts a force $\mathbf{F} = q\mathbf{E}_\sigma$ on the pointlike charge. The field is trivial to compute since it is equivalent to the one produced by the pointlike image charge. We find

$$|\mathbf{F}| = \frac{1}{4\pi\epsilon_0} \frac{qq'}{(y-y')^2} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2} \left(\frac{a}{y} \right)^3 \left(1 - \frac{a^2}{y^2} \right)^{-2}.$$

Consider two limits:

(i) When $y \gg a$

$$|\mathbf{F}| \sim y^{-3},$$

which is not Coulomb's law! This is the case because the sphere is grounded, meaning q' depends on y .

(ii) When $y \sim a$ the force becomes Coulomb-like

$$|\mathbf{F}| \sim (y-a)^{-2}.$$

This makes sense considering the induced charge distribution becomes highly peaked whenever q is close to the sphere.²²

2.3.2. Charged insulated conducting sphere

Consider a spherical conductor as before, but now we take it to be insulated (as opposed to grounded). This implies that the charge $Q \neq 0$ stored in the conductor is conserved, although it can still move around on S to achieve $\mathbf{E}_\parallel = 0$.

²²The distribution resembles a pointlike charge.

The solution to this can be found from the grounded case. Starting with a grounded sphere, then the presence of a pointlike charge q at $\mathbf{y}$ induces a charge q' over S as before. Now, we disconnect the conductor from ground, and add charge $Q - q'$ to the conductor. The total charge on the conductor is then $q' + (Q - q') = Q$ as required. The charge $Q - q'$ will distribute itself uniformly over S , since the induced q' on the sphere is already in equilibrium with q . We can therefore model the additional charge as a pointlike charge $Q - q'$ placed at the origin

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{|\mathbf{x} - \mathbf{y}|} - \frac{aq}{y|\mathbf{x} - \frac{a^2}{y^2}\mathbf{y}|} + \left(Q + \frac{a}{y}q \right) \frac{1}{|\mathbf{x}|} \right].$$

Likewise, the force on q is (probe approximation for q)

$$\mathbf{F} = q\mathbf{E}_{\text{sphere}} = \frac{1}{4\pi\epsilon_0} \frac{q}{y^2} \left[Q - \frac{qa^3(2y^2 - a^2)}{y(y^2 - a^2)^2} \right] \frac{\mathbf{y}}{y},$$

where $\mathbf{E}_{\text{sphere}}$ is sourced by the two image charges. There are some notable difference as compared to the grounded sphere:²³

- (i) When $y \gg a$ this agrees with Coulomb's law

$$|\mathbf{F}| \sim \frac{kqQ}{y^2},$$

since the sphere resembles a point charge Q .

- (ii) When $Qq < 0$ the force is always attractive.
- (iii) When $Qq > 0$ the force is repulsive beyond a certain distance. There is an unstable equilibrium y_* where $\mathbf{F} = 0$. The induced distribution (dipole-ish) dominates when $y \sim a$.

2.3.3. Conducting sphere in uniform electric field

Consider a spherical conductor of radius a in a uniform electric field $\mathbf{E} = (0, 0, E_0)$.

We model the background field by introducing pointlike charges $\pm Q$ placed at $z = \mp R$.

Taking R to be large, then this gives a field along the z -axis with

$$E_0 \simeq 2 \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}.$$

²³When $Q = 0$ some things change, importantly the Coulomb potential disappears.

Now, we place a conducting sphere of radius $a \ll R$ at the origin. The effect of the background can be computed using the method of images, since we model the field by pointlike charges²⁴

$$\pm Q \text{ at } \mathbf{y}_{\pm} = (0, 0, \mp R) \xrightarrow{\text{images}} \mp Q \frac{a}{R} \text{ at } \mathbf{y}'_{\pm} = \left(0, 0, \mp \frac{a^2}{R}\right).$$

We use coordinates $\mathbf{x} \sim (r, \theta)$, with r being the distance from the origin, and θ being the angle between $\mathbf{x}$ and the z -axis. The total potential is then the sum of our four charges

$$\Phi = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{(r^2 + R^2 + 2rR \cos \theta)^{1/2}} - \frac{Q}{(r^2 + R^2 - 2rR \cos \theta)^{1/2}} - \frac{aQ}{R \left(r^2 + \frac{a^4}{R^2} + \frac{2a^2r}{R} \cos \theta\right)^{1/2}} + \frac{aQ}{R \left(r^2 + \frac{a^4}{R^2} - \frac{2a^2r}{R} \cos \theta\right)^{1/2}} \right].$$

Our approximation where the background created by $\pm Q$ is uniform is only valid when $r \ll R$. Hence, we expand the above for $r \ll R$

$$\begin{aligned} \Phi &= -\frac{1}{4\pi\epsilon_0} \frac{2Q}{R^2} r \cos \theta \left(1 - \frac{a^3}{r^3}\right) + \dots \\ &= \underbrace{-E_0 r \cos \theta}_{\text{pure background field}} \overbrace{\left(1 - \frac{a^3}{r^3}\right)}^{\substack{\text{distortion from sphere} \\ \text{dipole-like}}} + \dots \end{aligned}$$

The distortion is caused by an induced charge density $\sigma(\mathbf{x})$ on the sphere

$$\sigma = \epsilon_0 \left(-\frac{\partial \Phi}{\partial n} \right)_{x=a} = 3\epsilon_0 E_0 \cos \theta.$$

This charge distribution is a dipole, since $\sigma \sim \cos \theta$. By symmetry (or Gauss' law) the total charge on the conductor vanishes, meaning an insulated sphere would lead to the same conclusions.²⁵

²⁴The q' and $\mathbf{y}'$ are analogous to the conducting sphere with a pointlike charge q .

²⁵Grounding the sphere does nothing, since no charges need to move from “ground” to the conductor in order to balance $\mathbf{E}$.

2.4. Green's function for the sphere

Consider a spherical surface of radius a centered at $\mathbf{x}_0 = \mathbf{0}$ with a given potential $\Phi_S(\theta, \phi)$ (on the surface). Suppose there is a charge distribution $\rho(\mathbf{x})$ outside the sphere. We would like to determine the potential outside the sphere: $|\mathbf{x} - \mathbf{x}_0| > a$, imposing Dirichlet boundary conditions $\Phi(|\mathbf{x} - \mathbf{x}_0| = a) = \Phi_S$. The missing piece is the Dirichlet Green's function G_D .

The defining equation for the Green's function is

$$\nabla'^2 G(\mathbf{x}, \mathbf{x}') = -4\pi\delta^{(3)}(\mathbf{x} - \mathbf{x}').$$

This says $G(\mathbf{x}, \mathbf{x}')$ is a solution to the Poisson equation with a pointlike source in the region V (outside the sphere in this case). The Green's function is simply the potential at $\mathbf{x}$ due to a pointlike source at $\mathbf{x}'$ (with suitable boundary conditions on some surface S). We require homogeneous Dirichlet boundary conditions: $G_D(\mathbf{x}, \mathbf{x}') = 0$ whenever $|\mathbf{x}| = a$.²⁶

Then solving for $G_D(\mathbf{x}, \mathbf{x}')$ is equivalent to computing the potential outside a grounded conducting sphere with a pointlike charge of magnitude $q = 4\pi\epsilon_0$ at $\mathbf{x}'$!²⁷ We have already solved this problem (our first example)

$$\begin{aligned} G_D(\mathbf{x}, \mathbf{x}') &= \frac{1}{|\mathbf{x} - \mathbf{x}'|} - \frac{a}{x' \left| \mathbf{x} - \frac{a^2}{x'^2} \mathbf{x}' \right|} \\ &= \frac{1}{(x^2 + x'^2 - 2xx' \cos \gamma)^{1/2}} - \frac{1}{\left[\left(\frac{xx'}{a} \right)^2 + a^2 - 2xx' \cos \gamma \right]^{1/2}}, \end{aligned}$$

²⁶This should be the case even if the potential has non-zero boundary values. The Poisson equation is a linear equation, meaning if Φ_1 and Φ_2 are solutions, then any linear combination of them is also a solution

$$\underbrace{\nabla'^2 \left[\sum_i q_i G_D(\mathbf{x}, \mathbf{x}'_i) \right]}_{\Phi} = -4\pi \underbrace{\sum_i q_i \delta^{(3)}(\mathbf{x} - \mathbf{x}'_i)}_{\rho}.$$

The total potential Φ satisfies the Dirichlet boundary condition if each Green's function does. Hence, the potential Φ with multiple sources is simply the sum of the Green's functions $G_D(\mathbf{x}, \mathbf{x}'_i)$ for each pointlike source. However, boundary conditions also add up. When $\Phi_i(S) = 0$ then their sum satisfies the same condition (obviously). This is why these Green's functions are useful. When $\Phi_i(S) = c$ the problem can be treated by adding an overall background constant.

²⁷Compare with

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0} \stackrel{\text{pointlike particle}}{=} -\frac{q}{\epsilon_0} \delta^{(3)}(\mathbf{x} - \mathbf{x}')$$

where γ is the angle between $\mathbf{x}$ and $\mathbf{x}'$.

Now, we can compute the potential using

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G_D(\mathbf{x}, \mathbf{x}') d^3x' - \frac{1}{4\pi} \oint_S \Phi_S(\mathbf{x}') \frac{\partial}{\partial n'} G_D(\mathbf{x}, \mathbf{x}') da'.$$

As a warmup, consider the case when $\Phi_S = 0$

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G_D(\mathbf{x}, \mathbf{x}') d^3x'.$$

One easily sees this potential satisfies (using the definition of G_D)

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0}, \quad \Phi_S = 0,$$

as desired.

Now, consider the more general Dirichlet boundary conditions

$$\Phi(|\mathbf{x}| = a) = \Phi_S(a, \theta, \phi).$$

We find

$$\Phi(\mathbf{x}) = \underbrace{\frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{x}') G_D(\mathbf{x}, \mathbf{x}') d^3x'}_{\text{potential in } V \text{ due to conductor at } S} - \underbrace{\frac{1}{4\pi} \oint_S \Phi_S(a, \theta', \phi') \frac{\partial}{\partial n'} G_D(\mathbf{x}, \mathbf{x}') da'}_{\text{correction term}},$$

which can be evaluated by noting²⁸

$$\left(\frac{\partial G_D}{\partial n'} \right)_{|\mathbf{x}'|=a} = -\frac{x^2 - a^2}{a(x^2 + a^2 - 2ax \cos \gamma)^{3/2}},$$

and $da' = a^2 d\Omega'$, where $d\Omega'$ is the solid angle at a point (a, θ', ϕ') and

$$\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi').$$

2.5. Separation of variables

Another technique we could use is separation of variables. However, this typically requires our problem has symmetry. The main idea is to exchange a single PDE in several variables to several (solvable) single-variable ODEs. This being possible depends on the given PDE, however, the Laplace equation has this property.

The Laplace equation is (in Cartesian coordinates)

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0.$$

²⁸This becomes a delta function (ish) in θ and ϕ as $|\mathbf{x}| \rightarrow a$.

There is a class of solutions in the form

$$\Phi(x, y, z) = X(x)Y(y)Z(z),$$

for which the Laplace equation becomes

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0.$$

This should hold for arbitrary x, y, z implying

$$\frac{d^2 X}{dx^2} = -\alpha^2 X, \quad \frac{d^2 Y}{dy^2} = -\beta^2 Y, \quad \frac{d^2 Z}{dz^2} = \gamma^2 Z,$$

with $\alpha^2 + \beta^2 = \gamma^2$. The solution is given by

$$X = e^{\pm i\alpha x}, \quad Y = e^{\pm i\beta y}, \quad Z = e^{\pm i\gamma z},$$

with the allowed frequencies depending on the boundary conditions. This solution is special, however, it can be shown that the general solution to the Laplace equation is a linear combination of such “product solutions”.²⁹

2.5.1. 3D box with conducting boundaries

Consider a rectangular box of sides (a, b, c) , where the boundaries are conductors with $\Phi = 0$ except for the top one ($z = c$) which has

$$\Phi(x, y, c) = V(x, y),$$

for some arbitrary function $V(x, y)$.

Applying separation of variables, we find (using $\Phi(x = 0) = \Phi(x = a) = 0$ etc.)

$$X = \sin \alpha_n x, \quad \alpha_n = \frac{n\pi}{a}$$

$$Y = \sin \beta_m y, \quad \beta_m = \frac{m\pi}{b}$$

$$Z = \sinh \gamma_{nm} z = \sinh \left(\sqrt{\alpha_n^2 + \beta_m^2} z \right).$$

Then by linearity, a linear combination of these is also a solution. We demand this general linear combination, with coefficients A_{nm} , satisfies the final boundary condition

$$\Phi(x, y, c) = \sum_{n,m=1}^{\infty} A_{nm} \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} c) = V(x, y).$$

²⁹Orthonormality and completeness is a general property of higher-dimensional Sturm-Liouville problems (like the Laplace equation).

This is a Fourier series in (x, y) , meaning it can be inverted to identify $A_{nm} \sinh(\gamma_{nm}c)$ as the Fourier coefficients of $V(x, y)$

$$A_{nm} \sinh(\gamma_{nm}c) = \frac{2}{a} \frac{2}{b} \int_0^a dx \int_0^b dy V(x, y) \sin(\alpha_n x) \sin(\beta_m y).$$

When there is a non-trivial potential on each wall we can use linearity to only consider one non-trivial wall at a time, and then sum our solutions

$$\Phi = \Phi_{x=a} + \Phi_{x=0} + \dots$$

2.5.2. Corners and edges in 2D

Consider conducting plates meeting at an angle $0 < \beta < 2\pi$. Using polar coordinates (ρ, ϕ) we use the following boundary conditions for the potential (our plates)

$$\Phi(\rho \geq 0, 0) = \Phi(\rho \geq 0, \beta) = V.$$

The Laplace equation in polar coordinates is³⁰

$$\left[\left(\rho \frac{\partial}{\partial \rho} \right)^2 + \frac{1}{\rho^2} \frac{\partial^2}{\partial \phi^2} \right] \Phi = 0.$$

Applying separation of variables with $\Phi(\rho, \phi) = R(\rho)F(\phi)$ we obtain two independent ODEs

$$\frac{1}{R} \left(\rho \frac{d}{d\rho} \right)^2 R = \nu^2, \quad \frac{1}{F} \frac{d^2}{d\phi^2} F = -\nu^2.$$

The solutions are known

$$R(\rho) = a\rho^\nu + b\rho^{-\nu}, \quad F(\phi) = A \cos(\nu\phi) + B \sin(\nu\phi), \quad (\nu \neq 0)$$

$$R(\rho) = a_0 + b_0 \ln \rho, \quad F(\phi) = A_0 + B_0 \phi. \quad (\nu = 0)$$

The boundary conditions require

$$b_0 = B_0 = b = A = 0, \quad \nu = \frac{m\pi}{\beta}, \quad m = 1, 2, \dots$$

Hence, the general solution becomes

$$\begin{aligned} \Phi(\rho, \phi) &= V + \sum_{m \geq 1} a_m \rho^{m\pi/\beta} \sin\left(\frac{m\pi\phi}{\beta}\right) \\ &\stackrel{\text{near } \rho=0}{=} V + a_1 \rho^{\pi/\beta} \sin\left(\frac{\pi\phi}{\beta}\right) + \dots \end{aligned}$$

³⁰Can be seen by using $\rho^2 = x^2 + y^2$ and $\phi = \arctan \frac{y}{x}$.

We expand near $\rho = 0$ since we would like the behaviour near the vertex. Here, there is still an ambiguity $(a_1, a_2, \dots)$ because we have not imposed any boundary conditions away from the vertex.

Now, we can read off the components of the electric field

$$E_\rho = -\frac{\partial\Phi}{\partial\rho} = -\frac{\pi a_1}{\beta} \rho^{\pi/\beta-1} \sin\left(\frac{\pi\phi}{\beta}\right)$$

$$E_\phi = -\frac{1}{\rho} \frac{\partial\Phi}{\partial\phi} = -\frac{\pi a_1}{\beta} \rho^{\pi/\beta-1} \cos\left(\frac{\pi\phi}{\beta}\right),$$

and the charge distribution near the tip from $E_\perp$ at $\phi = 0$

$$\sigma(\rho) = \epsilon_0 E_\phi(\rho, 0) \simeq -\frac{\pi a_1}{\beta} \rho^{\pi/\beta-1}.$$

This depends on the angle:

- (i) When $\beta < \pi$ the charge density goes to zero (at the vertex).
- (ii) When $\beta = \pi$ the charge density is constant across the boundary (plane conductor).
- (iii) When $\beta > \pi$ the charge density goes to infinity (at the vertex).
- (iv) When $\beta \sim 2\pi$ the corner becomes a “stick”. The charge density grows maximally

$$\sigma(\rho) \sim \rho^{-1/2}.$$
³¹

2.6. Appendix: Deriving the Coulomb Green’s function

Without boundary conditions and with a single pointlike charge the Green’s function for the Laplacian is given by the Coulomb potential. We would like to show this (with $\mathbf{x}' = 0$)

$$\nabla^2 \frac{1}{r} = -4\pi\delta(r).$$

This holds in three spatial dimensions.

Observe that $\delta^{(3)}(\mathbf{x}) = \delta(r)$ is a distribution, defined by

$$\int_V \delta(r) = \begin{cases} 0 & \text{if } 0 \notin V \\ 1 & \text{if } 0 \in V \end{cases}$$

So we need to show

³¹This is why lighting rods work.

$$\int_V \nabla^2 \frac{1}{r} = \begin{cases} 0 & \text{if } 0 \notin V \\ -4\pi & \text{if } 0 \in V \end{cases}$$

Now, we have

$$\nabla^2 \frac{1}{r} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \frac{1}{r} \right) = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right),$$

which vanishes if $r \neq 0$. So we have shown the top line. To show the bottom line we apply the divergence theorem

$$\begin{aligned} \int_V \nabla^2 \frac{1}{r} &= \int_V \nabla \cdot \left(\nabla \frac{1}{r} \right) \\ &= \oint_{\partial V} \hat{\mathbf{r}} \cdot \left(\nabla \frac{1}{r} \right) da \\ &= \oint_{\partial V} \hat{\mathbf{r}} \cdot \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} \frac{1}{r} \right) da \\ &= \oint_{\partial V} \underbrace{\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}}_{=1} \underbrace{\left(\frac{\partial}{\partial r} \frac{1}{r} \right)}_{=-1} r^2 d\Omega \end{aligned}$$

$$\text{any surface enclosing origin has } \Omega=4\pi \quad \underline{\underline{=}} \quad -4\pi,$$

which shows the bottom line. This could also be used to show the top line. The final integral vanishes when $0 \notin V$ since $\Omega = 0$ in this case.

3. Spherical harmonics

3.1. The spherical Laplace equation

The Laplace equation in spherical coordinates is

$$\nabla^2 f = \frac{1}{r} \frac{\partial^2}{\partial r^2} (rf) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \stackrel{!}{=} 0.$$

Applying separation of variables with an ansatz

$$f(r, \theta, \phi) = \frac{U(r)}{r} P(\theta) Q(\phi),$$

and multiplying by $r^2 \sin^2 \theta / f$ we find

$$r^2 \sin^2 \theta \left[\frac{1}{U} \frac{d^2 U}{dr^2} + \frac{1}{P r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) \right] + \underbrace{\frac{1}{Q} \frac{d^2 Q}{d\phi^2}}_{-m^2} \stackrel{!}{=} 0.$$

The ϕ -dependence has been isolated, and we obtain the equation

$$\frac{d^2 Q}{d\phi^2} = -m^2 Q.$$

This has solutions

$$Q(\phi) = e^{\pm i m \phi},$$

and if Q is single-valued (i.e. well-defined), then m is an integer.

The other equation we find is

$$\underbrace{\frac{r^2}{U} \frac{d^2 U}{dr^2}}_{l(l+1)} + \frac{1}{P \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \stackrel{!}{=} 0,$$

which separates as

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) + \left[l(l+1) - \frac{m^2}{\sin^2 \theta} \right] P = 0$$

$$\frac{d^2 U}{dr^2} - \frac{l(l+1)}{r^2} U = 0.$$

The radial equation is solved by

$$U(r) = Ar^{l+1} + Br^{-l},$$

with l unknown.

3.2. The Legendre equation

The θ equation is usually written in terms of $x = \cos \theta$

$$\frac{d}{dx} \left[(1-x^2) \frac{dP}{dx} \right] + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P = 0,$$

which is called the generalised Legendre equation, and its solutions are the associated Legendre functions.

Consider the case $m^2 = 0$

$$\frac{d}{dx} \left[(1-x^2) \frac{dP}{dx} \right] + l(l+1)P = 0,$$

which is called the ordinary Legendre equation. We seek a single-valued and “nice” solution in the range $-1 \leq x \leq 1$.³² This can be found by assuming the solution is a power series of the form³³

$$P(x) = x^\alpha \sum_{j=0}^{\infty} a_j x^j.$$

The resulting polynomials are called the Legendre polynomials of order l .³⁴ These are typically represented by Rodrigues’ formula

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l.$$

These are orthogonal

$$\int_{-1}^1 P_{l'}(x) P_l(x) dx = \frac{2}{2l+1} \delta_{l'l},$$

and complete. This implies any function $f(x)$ can be represented by a Legendre series

$$f(x) = \sum_{l=0}^{\infty} A_l P_l(x).$$

For more details, see Jackson.

The solutions to the generalised Legendre equation ($m^2 \neq 0$) are the associated Legendre functions³⁵

³²Otherwise, the solution is unphysical.

³³Method of Frobenius.

³⁴Where $l = 0, 1, \dots$

³⁵Where $m = -l, \dots, 0, \dots, l$.

$$P_l^m(x) \stackrel{m \geq 0}{=} (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_l(x).$$

Using Rodrigues' formula we have

$$P_l^m(x) \stackrel{\text{any } m}{=} \frac{(-1)^m}{2^l l!} (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} (x^2-1)^l.$$

The generalised Legendre equation only depends on m^2 , meaning P_l^{-m} and P_l^m are proportional

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x).$$

These are orthogonal (in l)

$$\int_{-1}^1 P_{l'}^m(x) P_l^m(x) dx = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \delta_{l'l}.$$

3.3. The spherical harmonics

We have found our solution to the Laplace equation can be written as

$$f(r, \theta, \phi) = \frac{U(r)}{r} P_l^m(\cos \theta) Q_m(\phi).$$

Recall the $Q_m(\phi)$ are complete and orthogonal in m on $0 \leq \phi \leq 2\pi$, while the $P_l^m(\cos \theta)$ are complete and orthogonal in l (for each m) on $-1 \leq \cos \theta \leq 1$. This implies the $P_l^m Q_m$ are complete and orthogonal on the unit-sphere in l and m .

The normalised $P_l^m Q_m$ are called the spherical harmonics

$$Y_{lm}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi}.$$

Or, for negative m

$$Y_{l,-m}(\theta, \phi) = (-1)^m Y_{lm}^*(\theta, \phi).$$

These are orthogonal

$$\int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta Y_{l'm'}^* Y_{lm} = \delta_{l'l} \delta_{m'm},$$

and complete

$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) = \delta(\phi - \phi') \delta(\cos \theta - \cos \theta').$$

This implies any function $g(\theta, \phi)$ can be represented as

$$g(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} Y_{lm}(\theta, \phi),$$

where

$$A_{lm} = \int d\Omega Y_{lm}^*(\theta, \phi) g(\theta, \phi).$$

3.4. The addition theorem and Green's functions

Consider two points $\mathbf{x}$ and $\mathbf{x}'$, with an angle γ between them. Then

$$P_l(\cos \gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi),$$

which is the addition theorem, see Jackson for a proof.

Now, we would like an expansion of

$$G_D(\mathbf{x}, \mathbf{x}') = \frac{1}{|\mathbf{x} - \mathbf{x}'|}.$$

We will show

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \gamma).$$

We can rotate our axes until $\mathbf{x}'$ lies along the z -axis. Then the potential satisfies the Laplace equation (unless $\mathbf{x} = \mathbf{x}'$ where we find a δ -function), and has azimuthal symmetry ($m = 0$). This means we can expand the potential as

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}) P_l(\cos \gamma).$$

Take $\mathbf{x}$ to be on the z -axis then

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} \stackrel{\mathbf{x} \text{ on } z\text{-axis}}{=} \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-(l+1)}),$$

while

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \frac{1}{(r^2 + r'^2 - 2rr' \cos \gamma)^{1/2}} \rightarrow \frac{1}{|r - r'|},$$

which can be expanded

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \frac{1}{r_{>}} \sum_{l=0}^{\infty} \left(\frac{r_{<}}{r_{>}} \right)^l.$$

When $\mathbf{x}$ is off the z -axis we simply multiply by $P_l(\cos \gamma)$ (since the coefficients A_l and B_l are unchanged), and we are done.

Then by the addition theorem we obtain

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi),$$

which can be generalised using the method of images.

4. Multipoles and polarisation

4.1. The multipole expansion

We have only considered field configurations in the presence of charges and ideal conductors in empty space. This is unrealistic. We would instead like to describe the behaviour of fields in different types of material. This is made difficult by matter backreacting. We saw a similar phenomenon when considering the surface density of conducting spheres in the presence of a pointlike charge. However, when considering materials these anisotropies are on much smaller scales.

The anisotropies are organised by the multiple expansion, which can be treated as a hierarchy of contributions to the field:

- (i) (zeroth order): Electric monopole. This corresponds to the contribution of a single charge, which is spherically symmetric.
- (ii) (first order): Electric dipole. This can correspond to the contribution due to two charges of opposite values spaced apart. This has cylindrical symmetry.
- (iii) (higher orders): Electric multipoles. These transform in specific ways under rotations.

4.1.1. Pure electric dipole

We start with the simplest non-trivial configuration, namely the electric dipole. Consider two charges $\pm q$ spaced apart by a vector $\mathbf{d}$. The dipole moment is defined as³⁶

$$\mathbf{p} = q\mathbf{d}.$$

We compute its potential as

$$\Phi(\mathbf{x}) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\mathbf{x} - \mathbf{x}_+|} - \frac{1}{|\mathbf{x} - \mathbf{x}_-|} \right),$$

where $\mathbf{x}_\pm = \mathbf{x}_0 \pm \mathbf{d}/2$ with $\mathbf{x}_0$ being the position of the dipole by definition. Now, we expand in small $|\mathbf{d}|/|\mathbf{x}|$

$$\Phi(\mathbf{x}) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\mathbf{x} - \mathbf{x}_0 - \frac{\mathbf{d}}{2}|} - \frac{1}{|\mathbf{x} - \mathbf{x}_0 + \frac{\mathbf{d}}{2}|} \right)$$

³⁶Taking $|\mathbf{d}| \rightarrow 0$ and $q \rightarrow \infty$, while keeping $|\mathbf{p}|$ fixed we find a “pointlike dipole”.

$$\begin{aligned}
&= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\mathbf{x} - \mathbf{x}_0|} + \left(-\frac{\mathbf{d}}{2} \right) \cdot \nabla \frac{1}{|\mathbf{x} - \mathbf{x}_0|} + \dots + \text{the same} \right) \\
&= \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|^3} + \dots \\
&= -\frac{1}{4\pi\epsilon_0} \mathbf{p} \cdot \nabla \frac{1}{|\mathbf{x} - \mathbf{x}_0|} + \dots
\end{aligned}$$

Taking $|\mathbf{d}|/|\mathbf{x}| \rightarrow 0$ all higher order terms die, and we find the dipole potential

$$\Phi_{\text{dipole}} = -\frac{1}{4\pi\epsilon_0} \mathbf{p} \cdot \nabla \frac{1}{|\mathbf{x} - \mathbf{x}_0|}.$$

The corresponding electric field is

$$\begin{aligned}
\mathbf{E}_{\text{dipole}} &= -\nabla \Phi \\
&= -\frac{1}{4\pi\epsilon_0} \nabla \left[\frac{\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|^3} \right] \\
&= -\frac{1}{4\pi\epsilon_0} \left[\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0) \nabla \frac{1}{|\mathbf{x} - \mathbf{x}_0|^3} + \frac{1}{|\mathbf{x} - \mathbf{x}_0|^3} \nabla (\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0)) \right] \\
&= -\frac{1}{4\pi\epsilon_0} \left[\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0) \left(-\frac{3(\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|^5} \right) + \frac{\mathbf{p}}{|\mathbf{x} - \mathbf{x}_0|^3} \right] \\
&= \frac{1}{4\pi\epsilon_0} \frac{1}{|\mathbf{x} - \mathbf{x}_0|^3} \left[3 \frac{(\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} \frac{\mathbf{p} \cdot (\mathbf{x} - \mathbf{x}_0)}{|\mathbf{x} - \mathbf{x}_0|} - \mathbf{p} \right] \\
&= \frac{3\mathbf{n}(\mathbf{p} \cdot \mathbf{n}) - \mathbf{p}}{4\pi\epsilon_0 |\mathbf{x} - \mathbf{x}_0|^3},
\end{aligned}$$

where

$$\mathbf{n} = \frac{\mathbf{x} - \mathbf{x}_0}{|\mathbf{x} - \mathbf{x}_0|}.$$

Using polar coordinates (r, θ, z) with $\mathbf{n} = \hat{\mathbf{r}}$ and $\mathbf{p} = p\hat{\mathbf{z}}$ we have³⁷

$$\begin{aligned}
\Phi_{\text{dipole}} &= \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^3} \\
\mathbf{E}_{\text{dipole}} &= \frac{2p \cos \theta}{4\pi\epsilon_0 r^3} \hat{\mathbf{r}} + \frac{p \sin \theta}{4\pi\epsilon_0 r^3} \hat{\boldsymbol{\theta}}.
\end{aligned}$$

Hence, $E_\phi = 0$ implying cylindrical symmetry.

³⁷With $\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}$.

4.1.2. The expansion

When a material is effected by an external field $\mathbf{E}$, the charges inside move and rearrange on a microscopic scale. This will change the electric field from the material. When viewed from far away the distribution “collapses” to a single point, and these contributions resemble a superposition of multipoles.

Suppose we have a charge distribution $\rho(\mathbf{x})$ confined in a region of radius R

$$\rho(\mathbf{x}) = 0, \text{ if } |\mathbf{x}| > R.$$

Outside this ball of radius R the potential satisfies the Laplace equation. Hence, we can write

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l (f_{l,m} r^l + g_{l,m} r^{-l-1}) Y_l^m(\theta, \phi).$$

This expression provides the basis for the multipole expansion of Φ . With $f_{l,m}$ and $g_{l,m}$ known we are done!

By Coulomb’s formula we know Φ is related to ρ by

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'.$$

Using the expansion for $1/|\mathbf{x} - \mathbf{x}'|$ we find (with $|\mathbf{x}'|$ smaller)

$$\begin{aligned} \Phi(\mathbf{x}) &= \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{x}') \left[4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r'^l}{r^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \right] d^3x' \\ &= \frac{1}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \left[\int Y_{lm}^*(\theta', \phi') r'^l \rho(\mathbf{x}') d^3x' \right] \frac{Y_{lm}(\theta, \phi)}{r^{l+1}} \\ &= \frac{1}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{q_{lm}}{2l+1} \frac{Y_{lm}(\theta, \phi)}{r^{l+1}}, \end{aligned}$$

which is the multipole expansion!

- (i) All $f_{l,m}$ vanish. This is expected since the potential decays as $r \rightarrow \infty$.
- (ii) There is no dependence on R , all information about ρ is carried by q_{lm} .
- (iii) The entire charge distribution is “squeezed at $r = 0$ ”, meaning $\Phi = \Phi(r, \theta, \phi)$.

We find the monopole moment ($l = 0$) is the electric charge

$$q_{00} = \frac{1}{\sqrt{4\pi}} \int \rho(\mathbf{x}') d^3x' = \frac{q}{\sqrt{4\pi}},$$

while the dipole moments ($l = 1$) corresponds to components of the electric dipole $\mathbf{p}$

$$q_{10} = \sqrt{\frac{3}{4\pi}} p_z$$

$$q_{1\pm 1} = -\sqrt{\frac{3}{8\pi}} (p_x \mp i p_y).$$

The multipole expansion becomes

$$\Phi(\mathbf{x}) = \frac{1}{\epsilon_0} \left[\underbrace{\frac{q_{00} Y_{00}}{r}}_{\text{pointlike charge}} + \underbrace{\frac{1}{3} \frac{q_{10} Y_{10} + q_{11} Y_{11} + q_{1-1} Y_{1-1}}{r^2}}_{\text{pointlike dipole}} + \dots \right]$$

$$= \Phi_{\text{monopole}} + \Phi_{\text{dipole}} + \Phi_{\text{quadropole}} + \dots$$

- (i) The multipole moments q_{lm} depend on the choice of origin. Placing a single pointlike charge at $r_0 = 0$ would only have $q_{00} \neq 0$, while placing the charge at $r_0 > 0$ will have infinitely many non-vanishing q_{lm} . This is due to symmetry.³⁸
- (ii) The multipole expansion can also be written in Cartesian coordinates

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon} \left[\frac{q}{r} + \frac{\mathbf{p} \cdot \mathbf{x}}{r^3} + \frac{1}{2} \sum_{i,j} Q_{ij} \frac{x_i x_j}{r^5} + \dots \right],$$

where (usually taken as a definition)

$$\mathbf{p} = \int \mathbf{x}' \rho(\mathbf{x}') d^3 x',$$

is the electric dipole moment of the charge distribution ρ , and

$$Q_{ij} = \int (3x'_i x'_j - r'^2 \delta_{ij}) \rho(\mathbf{x}') d^3 x',$$

is the electric quadropole tensor. Clearly Q_{ij} has five independent components.

4.2. Polarisation

Electric fields in materials are complicated. However, the dominant effects usually come from:

- (i) The “monopole density” $\rho(\mathbf{x})$ (usually vanishing).
- (ii) The “dipole density” $\mathbf{P}(\mathbf{x})$. This arises because external fields cause negative- and positive particles to get displaced in opposite directions, meaning we achieve a distribution of microscopic dipoles. This is polarisation!

³⁸Just use $\rho \sim \delta^{(3)}$ as a check. Here, $r'^l = 0$ for all $l > 0$.

The potential in a medium with only these is ($q \rightarrow \rho$ and $\mathbf{p} \rightarrow \mathbf{P}$ ³⁹)

$$\begin{aligned}\Phi(\mathbf{x}) &= \frac{1}{4\pi\epsilon_0} \int d^3x' \left[\frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} + \mathbf{P}(\mathbf{x}') \cdot \nabla' \frac{1}{|\mathbf{x} - \mathbf{x}'|} \right] \\ &\stackrel{\text{integration by parts}}{=} \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{|\mathbf{x} - \mathbf{x}'|} [\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')] \\ &= \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{\rho_{\text{eff}}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}.\end{aligned}$$

We have managed to rewrite the potential in Coulomb form! (with the sacrifice being $\rho \rightarrow \rho_{\text{eff}}$)

Applying ∇^2 we find the polarisation-corrected Poisson equation

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}(\rho - \nabla \cdot \mathbf{P}),$$

which can be written in terms of the electric displacement $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$

$$\nabla \cdot \mathbf{D} = \rho,$$

where ρ is the free charge! The idea is to combine the external field $\mathbf{E}$ with the polarisation field $\mathbf{P}$ caused by the response of the medium. The displacement $\mathbf{D}$ is necessarily less fundamental than the field $\mathbf{E}$, since everything microscopically is fields $\mathbf{E}$. However, on macroscopic scales all the microscopic details are averaged, and all the small changes in $\mathbf{E}$ due to non-trivial “micro-dynamics” leads to an effective description in terms of pointlike multipoles. The monopole moment is the zeroth order contribution, while the dipole moment (responsible for $\mathbf{P}$) is the first order contribution, and so on. The addition of these contributions is the macroscopic $\mathbf{D}$.

4.3. Electrostatics in macroscopic media

We have found the laws of electrostatic in macroscopic media are

$$\nabla \cdot \mathbf{D} = \rho, \quad \nabla \times \mathbf{E} = 0.$$

However, since we made no assumptions regarding the polarisation density, these laws are extremely general.

These imply the following⁴⁰

$$(\mathbf{D}_2 - \mathbf{D}_1) \cdot \mathbf{n} = \sigma$$

³⁹Compare with the Cartesian expansion.

⁴⁰The proof is identical to the case with $\mathbf{P} = 0$.

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0,$$

which generalises our previous behaviour for $\mathbf{E}$ at interfaces.

4.4. Linear dielectrics

With weak fields one can assume the polarisation response is linear

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E},$$

where χ_e is the electric susceptibility. Then the displacement is itself proportional to the field

$$\mathbf{D} = \epsilon \mathbf{E}, \quad \epsilon = \epsilon_0 \underbrace{(1 + \chi_e)}_{\text{dielectric constant}},$$

and a generalisation of Gauss' law holds in dielectrics

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon}.$$

This is huge, since everything we have done so far still holds (upon $\epsilon_0 \rightarrow \epsilon$).

We have e.g.

$$\begin{aligned} U &= \frac{1}{2} \int \rho \Phi \, d^3x' \\ &= \frac{1}{2} \int (\nabla \cdot \mathbf{D}) \Phi \, d^3x' \\ &= -\frac{1}{2} \int \mathbf{D} \cdot (\nabla \Phi) \, d^3x' \\ &= \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} \, d^3x' \\ &= \frac{\epsilon}{2} \int |\mathbf{E}|^2 \, d^3x', \end{aligned}$$

which differs from our previous result by an overall factor of ϵ/ϵ_0 . This occurs since energy is stored by $\mathbf{P}$.

We are now done with electrostatics, yay!

5. Magnetostatics

We have known for a long time that magnetic fields are sourced by electric charges in motion. However, a corresponding “magnetic charge” has never been found! This fundamental observation is what underlies all differences between electro- and magnetostatics.

5.1. Current and charge conservation

We define a current density $\mathbf{J}$, which is related to the current through a surface S as

$$I(S) = \frac{dq(S)}{dt} = \int_S \mathbf{n} \cdot \mathbf{J} \, da = \varphi_S(\mathbf{J}).$$

When the surface is closed, and S bounds a volume V , then the current measures how much charge leaves V . Also, by the divergence theorem we can write

$$\stackrel{\text{leaving } V}{=} \frac{d}{dt} \underbrace{\int_V \rho(\mathbf{x}) \, d^3x}_{q \text{ in } V} = \int_V \nabla \cdot \mathbf{J} \, d^3x.$$

This holds for arbitrary V implying

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0,$$

which is the continuity equation. By definition of magnetostatics (ρ time-independent), we find

$$\nabla \cdot \mathbf{J} = 0.$$

So we do allow moving charges if and only if there are no sinks or sources! Requiring no moving charges would be $\mathbf{J} \stackrel{!}{=} 0$.

5.2. Biot-Savart and Ampère’s laws

5.2.1. Biot-Savart’s law

Consider a (steady) $\mathbf{J}$ flowing in space. We imagine a small cylinder, of height dl and base area dS , which is aligned (i.e. parallel) with the flow lines of the current. We will denote the infinitesimal line element by $d\mathbf{l}$.

The current through dS is

$$I = \mathbf{J} \cdot \mathbf{n} \, dS = J \, dS,$$

and the contribution to the magnetic flux density $\mathbf{B}$ from the current through the cylinder $d\mathbf{l} \times dS$ is given by Biot-Savart's law

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{(I d\mathbf{l}) \times \mathbf{x}}{|\mathbf{x}|^3},$$

with the cylinder placed at $\mathbf{x} = 0$. This law resembles Coulomb's law, with some obvious differences.

5.2.2. Infinite wire

For an infinitely long and thin wire, the magnetic flux density at a distance R is given by

$$\begin{aligned} |\mathbf{B}| &= \frac{\mu_0}{4\pi} \int_{\mathbb{R}} \frac{|(I d\mathbf{l}) \times \mathbf{x}|}{|\mathbf{x}|^3} \\ &= \frac{\mu_0}{4\pi} \int_{\mathbb{R}} \frac{(I d\mathbf{l}) \times x \sin \theta}{x^3} \\ &= \frac{\mu_0}{4\pi} \int_{\mathbb{R}} \frac{IR}{(R^2 + l^2)^{3/2}} dl \\ &= \frac{\mu_0}{2\pi} \frac{I}{R}, \end{aligned}$$

with the direction given by the right-hand rule.

5.2.3. The vector potential

Computing $\mathbf{B}$ requires integrating the Biot-Savart law (as in the example) over all line elements. However, typically one deals with current distributions $\mathbf{J}$, describing charges moving in space. Here, $\mathbf{J}$ is a vector field over $\mathbb{R}^3$, and denoting the line elements along the flow lines of $\mathbf{J}$ by $d\mathbf{l}$ we can write

$$I d\mathbf{l} = \mathbf{J} dS \cdot d\mathbf{l} = \mathbf{J} d^3x,$$

implying

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}') \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} d^3x'.$$

We can rearrange this as

$$\begin{aligned} \mathbf{B}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \nabla \times \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\ &= \nabla \times \mathbf{A}(\mathbf{x}), \end{aligned}$$

with $\mathbf{A}$ being the magnetic vector potential.

The vector potential is unchanged under gauge transformations, since

$$\mathbf{A} \sim \mathbf{A} + \nabla \Psi,$$

lead to the same $\mathbf{B}$. Hence,

$$\mathbf{A} \neq \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'.$$

5.2.4. Ampère's law

Taking the divergence and curl of the equation for $\mathbf{B}$ we find the two fundamental equation of magnetostatics in vacuum⁴¹

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

The first is a trivial consequence of the existence of $\mathbf{A}$, and can be seen as the “Gauss’ law for magnetism”. This equation tells us there are no magnetic charges, since

$$q_m = \mu_0 \varphi_S(\mathbf{B}) = \mu_0 \int_V \nabla \cdot \mathbf{B} = 0.$$

The second equation is Ampère’s law, which plays the role of Gauss’ law, since it relates the field $\mathbf{B}$ to the sources. We can rewrite this equation by considering an open surface S with some boundary ∂S . Then integrating we find

$$\int_S (\nabla \times \mathbf{B}) \cdot \mathbf{n} da = \mu_0 \underbrace{\int_S \mathbf{J} \cdot \mathbf{n} da}_{I(S)},$$

and using Stokes’ theorem

$$\underbrace{\chi_{\partial S}(\mathbf{B})}_{\text{circulation of } \mathbf{B}} = \oint_{\partial S} \mathbf{B} \cdot d\mathbf{l} = \mu_0 I(S).$$

This relates the circulation of $\mathbf{B}$ to the current through the surface S , and usually provides a quicker way to compute $\mathbf{B}$.

We can also write Ampère’s law in terms of $\mathbf{A}$. Using the Coulomb gauge (i.e. gauge fixing)

$$\nabla \cdot \mathbf{A} = 0,$$

we have

$$\mu_0 \mathbf{J} = -\nabla^2 \mathbf{A}.$$

⁴¹Using $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$.

This tells us each A_i satisfies the Poisson equation, and therefore “acts like” the scalar potential. Hence, we may deduce

$$\mathbf{A}(\mathbf{x}) \stackrel{\text{Coulomb}}{=} \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'.$$

5.3. “Multipole expansion”

We can find a multipole expansion for $\mathbf{A}(\mathbf{x})$ as we did for the scalar potential. However, in this case things are annoying, since the A_i are coupled due to our choice of gauge. Hence, we will approach the problem differently.

Consider a localised (in small region) distribution $\mathbf{J}(\mathbf{x}')$, which will induce a field at $\mathbf{x}$. Assuming $|\mathbf{x}| \gg |\mathbf{x}'|$ we expand

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \frac{1}{|\mathbf{x}|} + \frac{\mathbf{x} \cdot \mathbf{x}'}{|\mathbf{x}|^3} + \dots$$

Then each A_i becomes

$$A_i(\mathbf{x}) = \frac{\mu_0}{4\pi} \left[\frac{1}{|\mathbf{x}|} \int J_i(\mathbf{x}') d^3x' + \frac{\mathbf{x}}{|\mathbf{x}|^3} \cdot \int J_i(\mathbf{x}') \mathbf{x}' d^3x' + \dots \right]$$

Since $\mathbf{J}(\mathbf{x}')$ is localised we have the following identity

$$\int (f \mathbf{J} \cdot \nabla' g + g \mathbf{J} \cdot \nabla' f + f g \nabla' \cdot \mathbf{J}) d^3x' = 0,$$

where f and g are “nice” functions of $\mathbf{x}'$. Using $f = 1$ and $g = x'_i$, we have

$$\begin{aligned} 0 &= \int \mathbf{J}(\mathbf{x}') \cdot \hat{\mathbf{x}}_i d^3x' \\ &= \int J_i(\mathbf{x}') d^3x'. \end{aligned}$$

Hence, the term corresponding to a monopole vanishes. With $f = x'_i$ and $g = x'_j$, we have

$$0 = \int x'_i J_j(\mathbf{x}') + x'_j J_i(\mathbf{x}') d^3x'.$$

The second term can then be written as

$$\begin{aligned} \mathbf{x} \cdot \int J_i \mathbf{x}' d^3x' &= \sum_j x_j \int x'_j J_i d^3x' \\ &= -\frac{1}{2} \sum_j x_j \int (x'_i J_j - x'_j J_i) d^3x' \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2} \sum_j \sum_k \epsilon_{ijk} x_j \int (\mathbf{x}' \times \mathbf{J})_k d^3x' \\
&= -\frac{1}{2} \left[\mathbf{x} \times \int (\mathbf{x}' \times \mathbf{J}) d^3x' \right]_i
\end{aligned}$$

One typically defines the magnetic dipole moment

$$\begin{aligned}
\mathbf{m} &= \frac{1}{2} \int \mathbf{x}' \times \mathbf{J} d^3x' \\
&= \frac{I}{2} \oint \mathbf{x}' \times d\mathbf{l}.
\end{aligned}$$

Then $\mathbf{A}(\mathbf{x})$ can be written as

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{x}}{|\mathbf{x}|^3}.$$

Taking the curl one immediately finds

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}) - \mathbf{m}}{|\mathbf{x}|^3}.$$

- (i) This is an approximation only valid when far away! Otherwise, one finds a multipole-like expansion.
- (ii) Viewing $\mathbf{J}$ as a vector field, each flow line can be regarded as an infinitesimally thin current loop. When a current loop is planar, the magnetic dipole is orthogonal to the plane, with size

$$\delta|\mathbf{m}| = \delta I \times \text{Area},$$

regardless of the loop shape. These loops can be used as a model for magnetic dipoles.

- (iii) For a distribution of charges with

$$\mathbf{J} = \sum_i q_i \mathbf{v}_i \delta^{(3)}(\mathbf{x} - \mathbf{x}_i)$$

we have

$$\mathbf{m} = \frac{1}{2} \sum_i q_i (\mathbf{x}_i \times \mathbf{v}_i) = \frac{1}{2} \sum_i \frac{q_i}{M_i} \mathbf{L}_i.$$

When all charges have the same q/M , then

$$\mathbf{m} = \frac{q}{2M} \mathbf{L},$$

implying the magnetic dipole moment is proportional to angular momentum!

5.4. Magnetic forces

Ampère observed that a wire carrying an electric current I in a magnetic flux density $\mathbf{B}$ is affected by a force

$$d\mathbf{F} = \underbrace{(I d\mathbf{l})}_{\text{probe current}} \times \mathbf{B},$$

which is the magnetic analogue to $\mathbf{F} = q\mathbf{E}$. Applying this to a point particle of charge q , where $I d\mathbf{l} = q\mathbf{v}$, we have

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B},$$

which is Lorentz' force law.

The force on a “probe distribution” due to a background field $\mathbf{B}$ (with $I d\mathbf{l} = \mathbf{J} d^3x$) is

$$\mathbf{F} = \int \underbrace{\mathbf{J} \times \mathbf{B}}_{\text{force density}} d^3x.$$

The torque density is defined by $\mathbf{x} \times (\mathbf{J} \times \mathbf{B})$. Hence, the overall torque is

$$\mathbf{N} = \int \mathbf{x} \times (\mathbf{J} \times \mathbf{B}) d^3x.$$

Assume the probe distribution is localised. We can expand a component of $\mathbf{B}$ as

$$B_k(\mathbf{x}) = B_k(0) + \mathbf{x} \cdot \nabla B_k(0) + \dots$$

Then F_i becomes

$$\begin{aligned} F_i &= \sum_{j,k} \epsilon_{ijk} \left[B_k(0) \underbrace{\int J_i(\mathbf{x}') d^3x'}_0 + \int J_j(\mathbf{x}') \mathbf{x}' \cdot \nabla B_k(0) d^3x' + \dots \right] \\ &\stackrel{\text{as before}}{=} \sum_{j,k} \epsilon_{ijk} (\mathbf{m} \times \nabla)_j B_k(\mathbf{x}), \end{aligned}$$

which can be written as (using $\nabla \cdot \mathbf{B} = 0$)

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B}).$$

The torque can similarly be found

$$\mathbf{N} = \int \mathbf{x}' \times [\mathbf{J} \times \mathbf{B}(0)] d^3x'$$

$$\begin{aligned}
&= \int \left[(\mathbf{x}' \cdot \mathbf{B})\mathbf{J} - \underbrace{(\mathbf{x}' \cdot \mathbf{J})\mathbf{B}}_0 \right] d^3x' \\
&= \mathbf{m} \times \mathbf{B}(0).
\end{aligned}$$

Now, we can define the potential energy a magnetic dipole

$$U = -\mathbf{m} \cdot \mathbf{B}.$$

5.5. Magnetisation

Whenever a field $\mathbf{B}$ propagates through a medium, there is a backreaction due to the interaction between $\mathbf{B}$ and electrons. This induces microscopic currents, whose macroscopic effect is producing anisotropies in the magnetic flux $\mathbf{B}$.

There are no monopoles, so the leading contribution in the multipole expansion comes from the magnetic dipole density or magnetisation $\mathbf{M}(\mathbf{x})$. The vector potential is then a superposition of the contribution from external $\mathbf{J}$ and the “internal” $\mathbf{M}$. Using the Coulomb gauge we have

$$\begin{aligned}
\mathbf{A}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int \left[\underbrace{\frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}}_{\text{current density}} + \underbrace{\frac{\mathbf{M}(\mathbf{x}') \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3}}_{\text{dipole density}} \right] d^3x' \\
&= \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} - \mathbf{M}(\mathbf{x}') \times \nabla' \frac{1}{|\mathbf{x} - \mathbf{x}'|} \right] d^3x' \\
&= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}') + \nabla' \times \mathbf{M}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\
&= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}_{\text{eff}}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x'.
\end{aligned}$$

Hence, we need to “change” Ampère’s law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_{\text{eff}},$$

with $\mathbf{J}_{\text{eff}} = \mathbf{J} + \nabla \times \mathbf{M}$. We can finally(!) define the magnetic field by

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M},$$

making Ampère’s law (with $\mathbf{J}$ free)

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

5.6. Magnetostatics in macroscopic media

We have found the laws of magnetostatics in macroscopic media are

$$\nabla \times \mathbf{H} = \mathbf{J}, \quad \nabla \cdot \mathbf{B} = 0.$$

Again, since we made no assumptions regarding the magnetisation, these laws are extremely general.

These imply the following

$$(\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{n} = 0$$

$$\mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K},$$

where $\mathbf{K}$ is the surface current density on the interface.

The magnetic energy density stored by the magnetic field in macroscopic media is

$$\begin{aligned} U &= \frac{1}{2} \int \mathbf{H} \cdot \mathbf{B} \, d^3x \\ &= \frac{1}{2} \int \mathbf{J} \cdot \mathbf{A} \, d^3x, \end{aligned}$$

which is also completely general.

5.6.1. Types of magnetisation

We have three broad classes of material responses to an external field $\mathbf{B}$:

- (i) Paramagnetism is characterised by a positive linear response

$$\mathbf{B} = \mu \mathbf{H}, \quad \frac{\mu}{\mu_0} > 1.$$

- (ii) Diamagnetism is characterised by a negative linear response

$$\mathbf{B} = \mu \mathbf{H}, \quad \frac{\mu}{\mu_0} < 1.$$

- (iii) Ferromagnetism is characterised by a nonlinear response

$$\mathbf{B} = \mathbf{F}(\mathbf{H}),$$

where $\mathbf{F}$ may be multivalued.

The parameter μ describing the linear response is known as magnetic permeability.

The usual laws apply to these materials after $\mu_0 \rightarrow \mu$. We can also define a magnetic susceptibility by

$$\mathbf{M} = \chi_m \mathbf{H},$$

implying $\mu = \mu_0(1 + \chi_m)$.

We are now done with magnetostatics, yay!

6. Maxwell's equations

6.1. Faraday's law

We have seen the following-ish:

$$\mathbf{E} \xrightarrow{\text{force}} q \text{ moves} \rightarrow \mathbf{J} \rightarrow \mathbf{B},$$

or briefly

$$\mathbf{E} \text{ induces } \mathbf{B}.$$

Consider a closed path C . We have seen

$$\oint_C \mathbf{E} \cdot d\mathbf{l} \stackrel{\text{electrostatics}}{=} 0.$$

This is related to the existence of a single-valued(!) potential Φ with

$$\mathbf{E} = -\nabla\Phi.$$

However, if the circulation is non-zero there will be a drop in potential

$$\mathcal{E} = \Delta\Phi(C) = \oint_C \mathbf{E} \cdot d\mathbf{l}.$$

This implies any charge placed on C will feel a force pushing it toward lower values of $q\Phi$. Hence, we call $\mathcal{E}$ the electromotive force.

The observed violation of the single-valuedness of Φ (in the presence of non-stationary currents⁴²) is the birth of electrodynamics! Faraday noticed a linear correlation between the $\mathcal{E}$ around C , and the change in $\varphi_S(\mathbf{B})$ across the surface S subtended by C

$$\begin{aligned} \mathcal{E} &= \oint_C \mathbf{E} \cdot d\mathbf{l} \\ &= -\frac{d}{dt} \int_S \mathbf{B} \cdot \mathbf{n} \, da \\ &= -\frac{d\varphi_S(\mathbf{B})}{dt}, \end{aligned}$$

which is Faraday's law. Note, the change in $\varphi_S(\mathbf{B})$ can be due to motion of C , time-dependence of $\mathbf{B}$, or both. Applying Stokes' theorem one finds

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

⁴²Here, both conditions defining electrostatics and magnetostatics are violated.

which is the differential form of Faraday's law.

One way to view Faraday's law is

$$\mathbf{B} \text{ induces } \mathbf{E}.$$

However, this is somewhat naïve.

6.2. Observer-dependence of Faraday's law

Suppose the path C is in motion with velocity $\mathbf{v}$ with respect to the lab, where a background field $\mathbf{B}$ is present. Then the electric field seen in the moving frame is (due to induction)

$$\begin{aligned} \oint_C \mathbf{E}_{\text{mov}} \cdot d\mathbf{l} &= -\frac{d}{dt} \int_S \mathbf{B} \cdot \mathbf{n} da \\ &\stackrel{S \text{ comoving}}{=} - \int_S \frac{d\mathbf{B}}{dt} \cdot \mathbf{n} da \\ &= - \int_S \left[\frac{\partial \mathbf{B}}{\partial t} + \underbrace{(\mathbf{v} \cdot \nabla) \mathbf{B}}_{\text{since } \mathbf{B} \text{ may be inhomogeneous}} \right] \cdot \mathbf{n} da \\ &= - \int_S \left[\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{B} \times \mathbf{v}) + \mathbf{v}(\nabla \cdot \mathbf{B}) \right] \cdot \mathbf{n} da. \end{aligned}$$

Applying $\nabla \cdot \mathbf{B} = 0$ and using Stokes' theorem we find

$$\oint_C [\mathbf{E}_{\text{mov}} + \mathbf{B} \times \mathbf{v}] \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} da,$$

which is Faraday's law in the moving frame.

An observer in the lab frame, however, sees

$$\oint_C \mathbf{E}_{\text{lab}} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot \mathbf{n} da,$$

since $\mathbf{B}$ only varies with time. Hence, we find

$$\mathbf{E}_{\text{mov}} = \mathbf{E}_{\text{lab}} + \mathbf{v} \times \mathbf{B}.$$

This is precisely Lorentz' force law upon multiplying with q ! This additional term is what breaks the single-valuedness of Φ , implying Coulomb's law is wrong.

6.3. Maxwell's equations

We know Faraday's law only plays a role when $\mathbf{B}$ is time-dependent. Therefore, induction as above only occurs in settings outside the domain of magnetostatics, where $\nabla \cdot \mathbf{J} \neq 0$. This follows immediately by considering the divergence of Ampère's law

$$\underbrace{\nabla \cdot (\nabla \times \mathbf{H})}_{\text{always 0}} = \nabla \cdot \mathbf{J} \neq 0.$$

Hence, we need to modify Ampère's law. The necessary modification simply turns out to be consistency with Gauss' law. Assuming $\nabla \cdot \mathbf{D} = \rho$ is still valid (i.e. taking Gauss' law as a definition), we find by charge conservation

$$\nabla \cdot \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \stackrel{!}{=} 0.$$

This means we can “fix” Ampère's law by addition of $\partial \mathbf{D} / \partial t$

$$\nabla \times \mathbf{H} = \mathbf{J} + \underbrace{\frac{\partial \mathbf{D}}{\partial t}}_{\substack{\text{displacement} \\ \text{current}}}.$$

This was proposed by Maxwell, and the equation is typically called Ampère-Maxwell's law. With this additional term we see a magnetic field can be induced by a time-varying electric field (even if no charges are moving!).

Now, we have found the four equations of electrodynamics or Maxwell's equations

$$\nabla \cdot \mathbf{D} = \rho \quad (\text{Gauss' law})$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{no monopoles})$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (\text{Ampère-Maxwell's law})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (\text{Faraday's law})$$

Alongside Lorentz' force law (on a single charged particle) these describe all phenomena in classical electrodynamics.

6.4. Potentials and wave equations

We already know

$$\mathbf{B} = \nabla \times \mathbf{A},$$

since $\nabla \cdot \mathbf{B} = 0$. Then by Faraday's law

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0,$$

implying

$$\mathbf{E} = \underbrace{-\nabla\Phi}_{\text{old definition}} - \frac{\partial \mathbf{A}}{\partial t}.$$

This is the “true” definition of the scalar potential in electrodynamics. This definition is again well-defined and single-valued everywhere!

These new potentials again have an intrinsic ambiguity, since

$$\mathbf{A} \sim \mathbf{A} + \nabla\Lambda, \quad \Phi \sim \Phi - \frac{\partial\Lambda}{\partial t},$$

leaves both $\mathbf{E}$ and $\mathbf{B}$ unchanged. The choice of $\Lambda(\mathbf{x}, t)$ is a redundancy in our description of the physical fields in terms of potentials. This is usually called gauge freedom.

Using $\mathbf{A}$ and Φ we immediately obtain the homogeneous Maxwell equations by definition. The inhomogeneous Maxwell equations become (in vacuum)

$$\nabla^2\Phi + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{\rho}{\epsilon_0}$$

$$\nabla^2\mathbf{A} - \frac{1}{c^2}\frac{\partial^2\mathbf{A}}{\partial t^2} - \nabla\left(\nabla \cdot \mathbf{A} + \frac{1}{c^2}\frac{\partial\Phi}{\partial t}\right) = -\mu_0\mathbf{J}.$$

Lorenz noticed these become nice if one applies gauge freedom to transform $\mathbf{A}$ and Φ by Λ with⁴³

$$\square\Lambda = -\left(\nabla \cdot \mathbf{A} + \frac{1}{c^2}\frac{\partial\Phi}{\partial t}\right).$$

This is called the Lorenz gauge, and the resulting potentials satisfy

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2}\frac{\partial\Phi}{\partial t} \stackrel{!}{=} 0.$$

This gauge is useful since the equations for $\mathbf{A}$ and Φ decouple, while simultaneously reducing both to wave equations

$$\square\Phi = -\frac{\rho}{\epsilon_0}$$

⁴³Here, $\square$ is the D’Alembertian

$$\square\Lambda = \nabla^2\Lambda - \frac{1}{c^2}\frac{\partial^2\Lambda}{\partial t^2}.$$

$$\square \mathbf{A} = -\mu_0 \mathbf{J}.$$

Then solving Maxwell's equation has been reduced to solving four copies of the wave equation! However, the Lorenz gauge is still semi-ambiguous since

$$\Lambda \sim \Lambda + \delta\Lambda,$$

if $\square \delta\Lambda = 0$.

Note, the potentials obeying wave equations should not(!) be confused with the occurrence of electromagnetic waves.

Another useful gauge is (again) the Coulomb gauge

$$\nabla \cdot \mathbf{A} = 0.$$

This recovers Gauss' law (or the Poisson equation for Φ)

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0},$$

meaning Φ is determined by Coulomb's law (namesake). Now, this is understood to hold also for a time-dependent charge distribution

$$\Phi(\mathbf{x}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{x}', t)}{|\mathbf{x} - \mathbf{x}'|} d^3x'.$$

This is strange since it appears to say a change in ρ propagates immediately to Φ (over arbitrary distances!).

With no sources $\mathbf{J} = 0$ we find in the Coulomb gauge

$$\Phi = 0, \quad \square \mathbf{A} = 0,$$

while the physical fields are simply

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

6.5. Green's function for the D'Alembertian

Maxwell's equations in the Lorenz gauge have the form

$$\square \Psi = -4\pi f(\mathbf{x}, t).$$

We would like to determine the Green's function for $\square$

$$\square G(\mathbf{x}, t; \mathbf{x}', t') = -4\pi \delta^{(3)}(\mathbf{x} - \mathbf{x}') \delta(t - t'),$$

with “no boundary” conditions.⁴⁴

⁴⁴Neumann boundary conditions at infinity

By symmetry G can only depend on distances

$$G = G(R; \tau),$$

with $R = |\mathbf{x} - \mathbf{x}'|$ and $\tau = t - t'$.

We Fourier transform using

$$G(R, \tau) = \frac{1}{2\pi} \int G_k(R) e^{-i\omega\tau} d\omega$$

$$\delta(\tau) = \frac{1}{2\pi} \int e^{-i\omega\tau} d\omega,$$

where $\omega = ck$. Then the wave equation for modes with frequency ω becomes⁴⁵

$$(\nabla^2 + k^2)G_k(R) = -4\pi\delta^{(3)}(\mathbf{R}),$$

which defines $G_k(R)$. The solutions are⁴⁶

$$G_k^{(\pm)}(R) = \frac{1}{R} e^{\pm ikR}.$$

Then transforming back we find

$$\begin{aligned} G^{(\pm)}(\mathbf{R}, \tau) &= \frac{1}{2\pi} \int G_k^{(\pm)}(R) e^{-i\omega\tau} d\omega \\ &= \frac{1}{2\pi} \int \frac{1}{R} e^{i(\pm kR - \omega\tau)} d\omega \\ &= \frac{1}{R} \delta\left(\pm \frac{R}{c} - \tau\right) \\ &= \frac{1}{|\mathbf{x} - \mathbf{x}'|} \delta\left(t - t' \mp \frac{|\mathbf{x} - \mathbf{x}'|}{c}\right). \end{aligned}$$

- (i) By definition, then these Green's functions describe the potentials Φ or $\mathbf{A}$ at the point $(\mathbf{x}, t)$ if a source ρ or $\mathbf{J}$ is turned on for an instant(!) at a point $(\mathbf{x}', t')$.
- (ii) Correlation is non-trivial when wave propagation occurs with the speed of light

$$G^{(\pm)}(\mathbf{R}, \tau) = 0, \quad \text{if } |\Delta\mathbf{x}| \neq c\delta t.$$

- (iii) Since

$$\frac{|\mathbf{x} - \mathbf{x}'|}{c} > 0,$$

⁴⁵This is simply the Helmholtz equation (with a δ -source).

⁴⁶Note, when $k = 0$ we recover the Poisson equation for a pointlike particle. This is also similar to the Yukawa potential.

we have

$$G^{(+)}(\mathbf{R}, \tau) = 0, \quad \text{if } t < t'.$$

There is no signal propagation for times t before a source at t' , only afterwards!

This is why $G^{(+)}$ is called the retarded Green's function (what happens after the impulse?).

(iv) Likewise

$$G^{(-)}(\mathbf{R}, \tau) = 0, \quad \text{if } t > t'.$$

There is no signal propagation for times t after a source at t' , only before. This

is why $G^{(-)}$ is called the advanced Green's function (what happens before the impulse?).

To see how this works consider a localised source $f(\mathbf{x}, t)$. We assume

$$f(\mathbf{x}, t) = 0, \quad t < t_{\text{in}} \text{ or } t > t_{\text{out}},$$

meaning $f(\mathbf{x}, t)$ is “on” between t_{in} and t_{out} . Suppose we know the solution $\Psi_{\text{in}}(\mathbf{x}, t)$ of $\square\Psi_{\text{in}} = 0$ before the source is activated. Then the overall solution is

$$\Psi(\mathbf{x}, t) = \Psi_{\text{in}}(\mathbf{x}, t) + \int d^3x' \int dt' G^{(+)}(\mathbf{x}, t; \mathbf{x}', t') f(\mathbf{x}', t').$$

(i) This obeys the wave equation with sources for all $\mathbf{x}$ and t

$$\square\Psi(\mathbf{x}, t) = -4\pi f(\mathbf{x}, t), \quad t \geq t_{\text{in}},$$

which is obvious (by definition).

(ii) When $t < t_{\text{in}}$ we have $\Psi(\mathbf{x}, t) = \Psi_{\text{in}}$ as we would expect.

(iii) However, after activation we find

$$\begin{aligned} \Psi(\mathbf{x}, t) &= \Psi_{\text{in}} + \int d^3x' \int dt' G^{(+)}(\mathbf{x}, t; \mathbf{x}', t') f(\mathbf{x}', t') \\ &= \Psi_{\text{in}} + \int d^3x' \int dt' \frac{1}{|\mathbf{x} - \mathbf{x}'|} \delta\left(t - t' - \frac{|\mathbf{x} - \mathbf{x}'|}{c}\right) f(\mathbf{x}', t') \\ &= \Psi_{\text{in}} + \int d^3x' \frac{1}{|\mathbf{x} - \mathbf{x}'|} f\left(\mathbf{x}', t - \frac{|\mathbf{x} - \mathbf{x}'|}{c}\right). \end{aligned}$$

We add contributions from the source evaluated at points $\mathbf{x}'$ at a previous(!) time $t' = t - |\Delta\mathbf{x}|/c$ corresponding to the time taken for the signal to propagate to $\mathbf{x}$. We evaluate the source using the “retarded time”.

- (iv) The strength of the signal decays as $1/|\mathbf{x} - \mathbf{x}'|$. This is the same behaviour as Coulomb's law for Φ (and Biot-Savart's law for $\mathbf{A}$). This means the Green's function tells us the potentials (and fields) take time to propagate, and a change in sources causes a change in the fields propagating with the speed of light.

A similar expression with Φ_{out} and $G^{(-)}$ can also be found.

When doing electrodynamics in the Lorenz gauge, we are (usually) interested in situations where $\Psi_{\text{in}} = 0$. Then applying the above we find

$$\Phi(\mathbf{x}, t) = \frac{1}{4\pi\epsilon} \int d^3x' \frac{1}{R} [\rho(\mathbf{x}', t')]_{\text{ret}}$$

$$\mathbf{A}(\mathbf{x}, t) = \frac{\mu_0}{4\pi} \int d^3x' \frac{1}{R} [\mathbf{J}(\mathbf{x}', t')]_{\text{ret}},$$

where $[\dots]_{\text{ret}}$ means we evaluate the term with

$$t' = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c} = t - \frac{R}{c}.$$

6.6. Appendix: The Helmholtz equation

The Helmholtz equation is

$$(\nabla^2 + k^2)G_k(r) = -4\pi\delta^{(3)}(\mathbf{r}).$$

This equation is spherically symmetric (only depends on r)

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (rG(r)) + k^2 G_k(r) = -4\pi\delta(r).$$

Away from $r = 0$ we have

$$\frac{\partial^2}{\partial r^2} (rG(r)) + k^2 rG_k(r) = 0, \quad r > 0.$$

Hence, the function $f_k(r) = rG_k(r)$ is harmonic

$$\frac{\partial^2}{\partial r^2} f_k(r) = -k^2 f_k(r), \quad r > 0,$$

with solutions

$$f_k^{(\pm)}(r) = e^{\pm ikr}, \quad r > 0,$$

and we immediately find two solutions for G_k aswell

$$G_k^{(\pm)}(r) = \frac{e^{\pm ikr}}{r}, \quad r > 0.$$

Now, we need to consider $r = 0$. The $f_k^{(\pm)}$ are functions of kr , so in the limit $r \rightarrow 0$ all functions behave in the same way (for all values of k !). This means we are free to use any value of k , including $k = 0$. Then G_k becomes the Coulomb Green's function, and both $f_k^{(\pm)} \rightarrow \frac{1}{r}$ which is the Coulomb potential. We conclude our found solutions also hold for $r = 0$. Hence, for all r

$$G_k^{(\pm)}(r) = \frac{1}{r} e^{\pm ikr}.$$